

Anchored Head–Analytic Tail Gluing Criterion

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Abstract

We isolate the exact finite-head/analytic-tail gluing step needed to turn cloud-extracted trace control into a locally uniform relative determinant. The logarithmic error is the sum of a finite anchored head error, a quotient tail, and the deterministic target tail; exponentiation gives an explicit determinant bound. Applying the ledger to the current relaxed shell class finds no complete certificate: the anchored head fails at all 32 endpoints, the quotient tail is certified only on 17 finite matrices, and the target tail is unbounded. This is a scoped route stop, not a global impossibility claim.

1 Exact gluing estimate

Let $\tau_n(\sigma)$ be a residual trace sequence and a_n a deterministic target. For R inside both logarithmic convergence domains, define

$$L_\sigma(z) = - \sum_{n \geq 2} \frac{\tau_n(\sigma)}{n} z^n, \quad L_*(z) = - \sum_{n \geq 2} \frac{a_n}{n} z^n.$$

For a finite head through N put

$$H_N(\sigma, R) = \sum_{n=2}^N \frac{|\tau_n(\sigma) - a_n| R^n}{n},$$

and let $T_\sigma(N, R)$ and $T_*(N, R)$ be the analogous absolute tails from $n > N$.

Theorem 1 (head–tail gluing). *For every $|z| \leq R$,*

$$|L_\sigma(z) - L_*(z)| \leq H_N(\sigma, R) + T_\sigma(N, R) + T_*(N, R). \quad (1)$$

If additionally $|L_\sigma(z)|, |L_(z)| \leq B$ on the disk, then for $D_\sigma = e^{L_\sigma}$ and $D_* = e^{L_*}$,*

$$|D_\sigma(z) - D_*(z)| \leq e^B (H_N + T_\sigma + T_*). \quad (2)$$

Uniform convergence follows if the three budgets tend uniformly to zero along a sequence of heads and radii.

Proof. Subtract the two absolutely convergent series and apply the triangle inequality separately to the finite head and the two tails. For the second claim, integrate the derivative of e^u along the line segment between the two logarithms; its modulus is at most e^B . $\square$

RH-246 supplies a conditional bound for T_σ from a contractive block:

$$T_\sigma(N, R) \leq \frac{K_m R^m}{m(1 - \eta_m R^m)} \sum_{r=0}^{m-1} L_r R^r$$

whenever $N \geq m - 1$; using a bound beginning at m is conservative for a tail beginning at $m + 1$ [3]. The target tail T_* is a separate coefficient theorem, not supplied by the finite RH-243 dictionary.

2 Current certificate ledger

We use the RH-248 convex shell relaxation for the head, so any failure also rules out the legal single-use shell class [2, 4]. The minimum and maximum anchored head distances are

$$0.14649763462315904 \quad \text{and} \quad 0.4240179027308174,$$

with zero passes at the 32 archived endpoints. The RH-246 12-block tail diagnostic gives $1.7991531976413385 \times 10^{-5}$ on only 17 endpoints. The smallest head distance is therefore 8142.588125081018 times this finite tail budget. The target tail is not bounded, and the two endpoint sets do not yet form a uniform family.

certificate	status
anchored head, 32 endpoints	0/32 passes
quotient tail, 17 finite endpoints	finite diagnostic only
deterministic target tail	not bounded
complete head/tail certificates	0

3 Boundary and route stop

Theorem 1 is exact and remains the correct interface to RH-240 [1]. The negative ledger is scoped to the frozen candidate class, order-12 head, unit disk, and finite subbatch used in RH-246. It does not exclude a new anchored cloud, an expanded window, a different function space, or a future target-tail theorem. It does establish that adding the current analytic-tail diagnostic cannot rescue the missing anchored head.

Gate A remains open and Gates B–E are untouched. No Hilbert–Polya operator, zeta-divisor equality, Riemann-zero identification, or RH implication is claimed.

References

- [1] Bin Wang. Uniform trace envelope criterion. 2026. RH-240 preprint.
- [2] Bin Wang. Deterministic numerator coefficient anchor dictionary. 2026. RH-243 preprint.
- [3] Bin Wang. Block-power quotient envelope criterion. 2026. RH-246 preprint.
- [4] Bin Wang. Anchored shell-zonotope reachability obstruction. 2026. RH-248 preprint.